

Weight Rack Design Report

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We were tasked with the designing of a weight rack with a handful of specifications. Namely, the rack itself could not weigh over 250 lbs and must be able to hold a total weight of 295 pounds in the form of four 45 pound plates, two 25 pound plates, four 10 pound plates, and five 5 pound plates. Beforehand we knew that we would have access to a variety of tools and machines including a manual mill, lathe, TIG welding machine, and a CNC machine. With all of these things in mind, I set out to design a lightweight, easy to build weight rack.

Upon first inspection, I thought that the best design would be the classic weight rack, with a design resembling a christmas tree, with a handful of pegs to hang the weight on. However as I thought more and more about this design I realized that there may be some better solutions out there. I soon realized that if I wanted to reduce the center of gravity, I should focus on a design that keeps the weights low to the ground and minimizes stress on the rack itself. This was when I decided to go with my final design that had the weight rack lay on the ground, designed to have the larger weights propped in between 3 sets of pipes in the middle of a square, and 2 pipes on the end of this rectangular frame to hold the smaller plates.

The first portion of my rack that I designed was the base plate, knowing that I was going to use steel, I wanted to reduce the weight by designing the base plate to have a hole in its center. I also drilled 4 holes in the corner of the plate so that when installing the rack, it could bolt directly into the ground and not move. Initially I dimensioned the base to be a 20"x10"x0.5"

rectangle but once I began the assembly I returned to the part and extended it to be a 25"x10"x0.5" rectangle to ensure that all the weights on the sides could be held.

Next I set off to design the pipes that I would attach to the corners of the base plate. Understanding that each plate was roughly 1" thick, I decided to make each of the pipes about 4 inches long with an outside diameter of 1.750" to ensure that the weights, which have an internal diameter of 2", would fit on nicely. Similarly to the base plate, I was forced to change my design upon assembly and the addition of weights, as I realized that having a pipe in every corner caused the weights to overlap when stacked on. So instead of lengthening the base plate anymore, I simply took a pipe off of each side and moved the remaining pipe to the center so that all the weights could fit on without overlap.

The last portion of the rack that I designed was the middle pipes that would hold up the larger weights. In order to keep the weight and center of gravity relatively low, I made them hollow pipes and gave them a total height of about 8". I chose to create them using a swept blend with a rather sharply pointed triangle, that way when ordering materials all that I'd have to do is cut two pipes at an angle and weld them together. For a while I couldn't decide if I wanted to have 5 holds or 3 holds, eventually deciding on 3 holds as 5 felt to be too many for such a small weight rack.

As for how everything is going to be held together, I decided to use fillet welds everywhere as they seemed to be best suited. Each of these welds would be done with a diameter of 0.250". As for the location of the welds, I'd have to weld the end pipes all around to the base plate, same goes for the triangle holds. On top of welding the triangle holds to the base, I'd need to weld the connection point of the two pipes of each triangle hold.

Once I had designed each individual part in CREO, I took them all into an assembly and began to construct the final weight rack. After making some adjustments mentioned earlier in the report, I added all of the weight and placed them in the places I desired. The next thing to do was run some stress analyses on my rack to ensure that nothing would break or bend too much when under max load.

Before running the analyses I made sure to create a skeleton model of my finished assembly that excluded the welds as we were told to assume that the welds would not fail before the parts. For the actual analysis, I only decided to run one since the end pipes would not be experiencing any stress due to the design of the base plate and end pipes. Instead I created an area along one of the triangle holds on which I applied a force of 115 lbf (2-45's and a 25 lb. plate). I assumed that since the weights inside would be leaning that their force would only be applied on the upper half of the triangle. After that I constrained the bottom of the base plate to simulate it being bolted to the floor.

As for the actual analysis itself, I ran both a max stress and displacement simulation and found the maximum stress to only be about 6.7 ksi, knowing that steel has a maximum yield stress of about 70 ksi this was more than acceptable. As for the deflection, I just needed to make sure that the deflection wasn't so much that the triangle hold would either hit the other plates or the floor. Again I found its max displacement to only be about 0.011564". Finally I decided to run a mass property measurement on the finalized product, finding the total weight to be about 24.682 lbs.

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Appendix

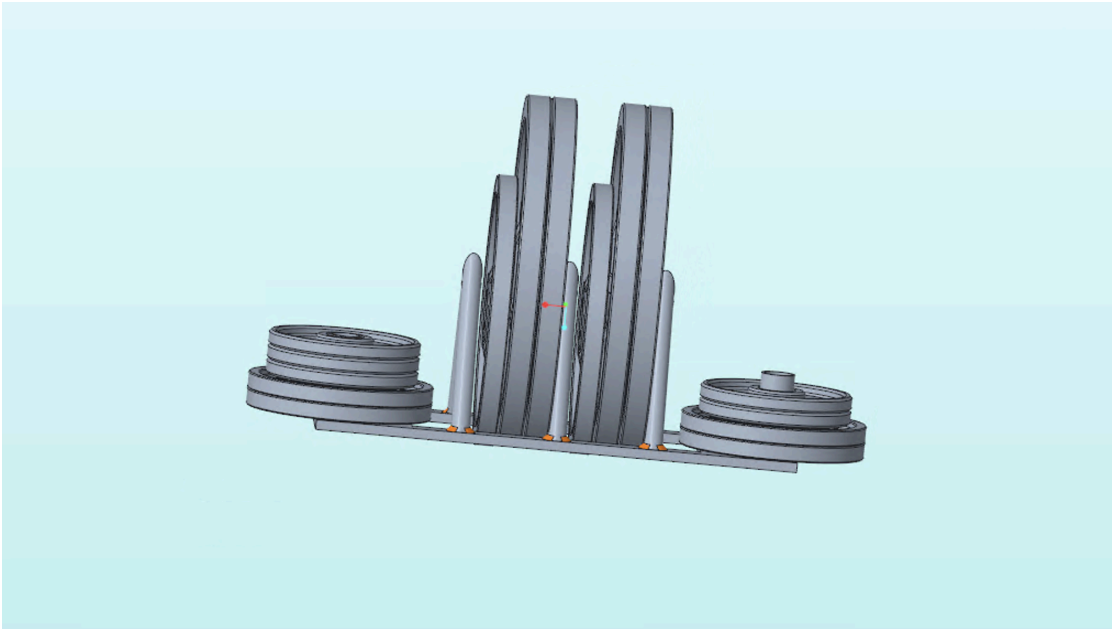


Figure 1: Fully Assembled Rack with all Weights

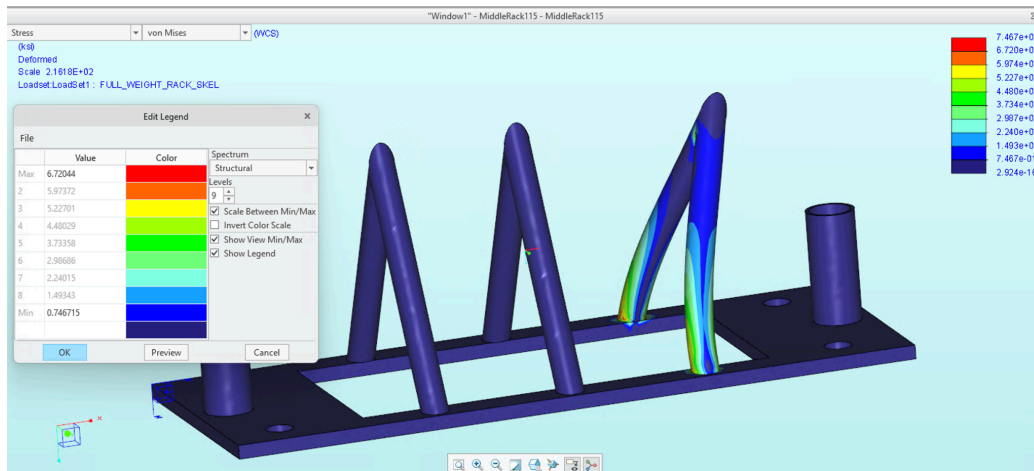


Figure 2: Stress Test w/ 115 lbf

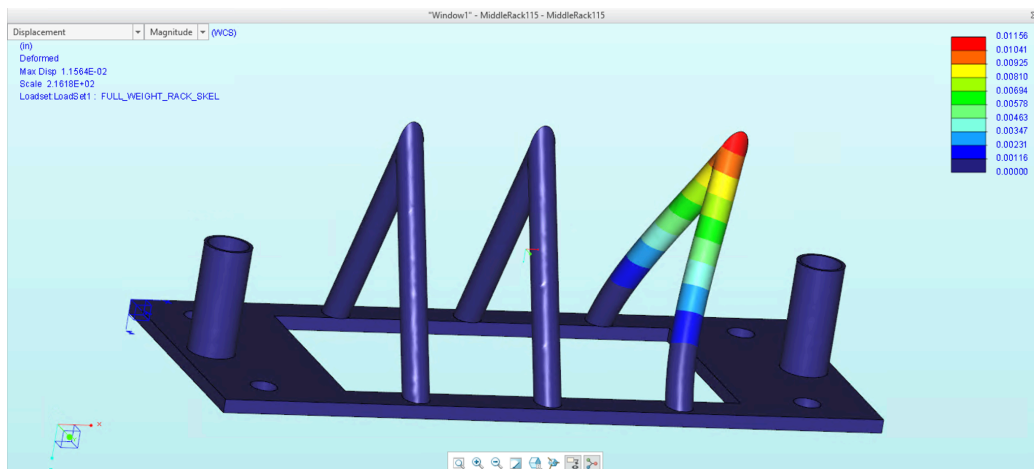


Figure 3: Displacement test w/ 115 lbf

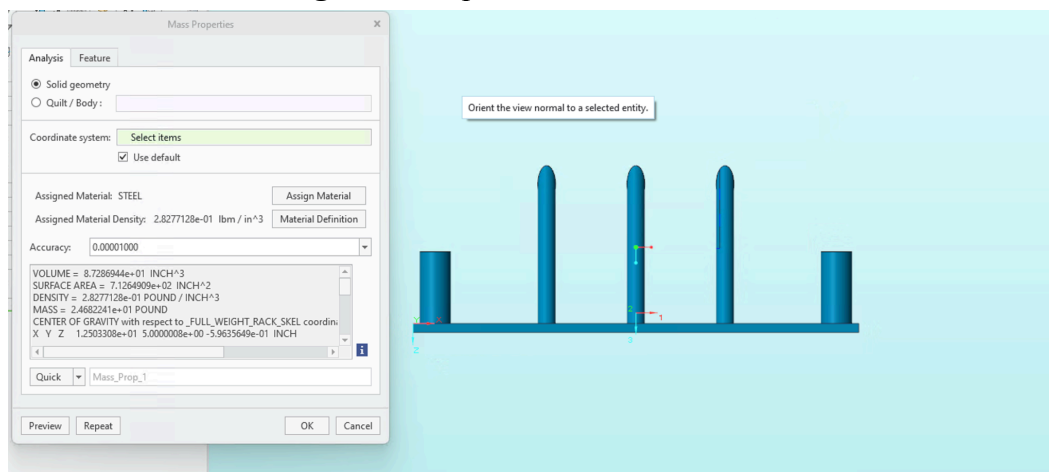


Figure 4: Center of Gravity and Mass

Center of Gravity: X Y Z 1.2503308e+01 5.0000008e+00 -5.9635649e-01 INCH

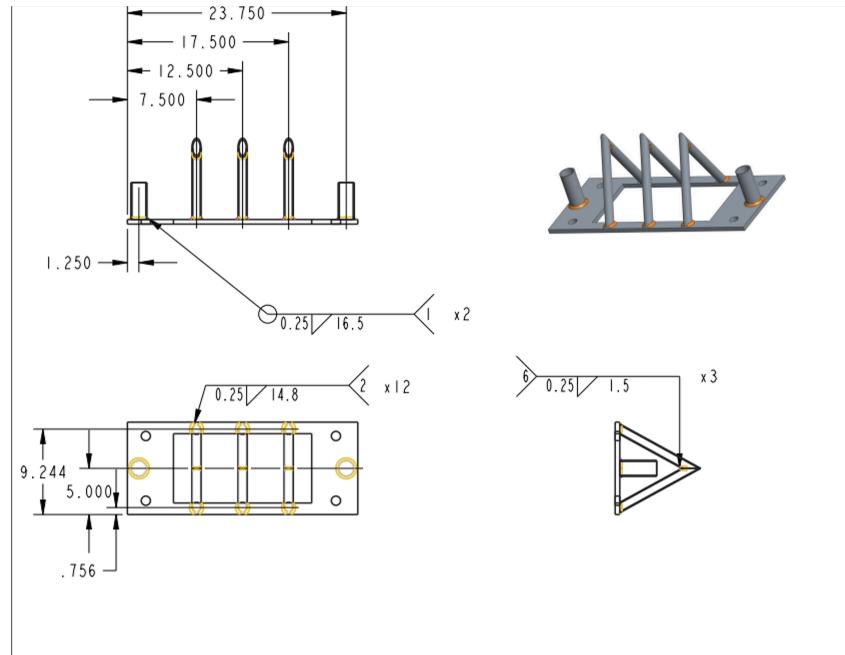


Figure 5: Fully Assembled Drawing

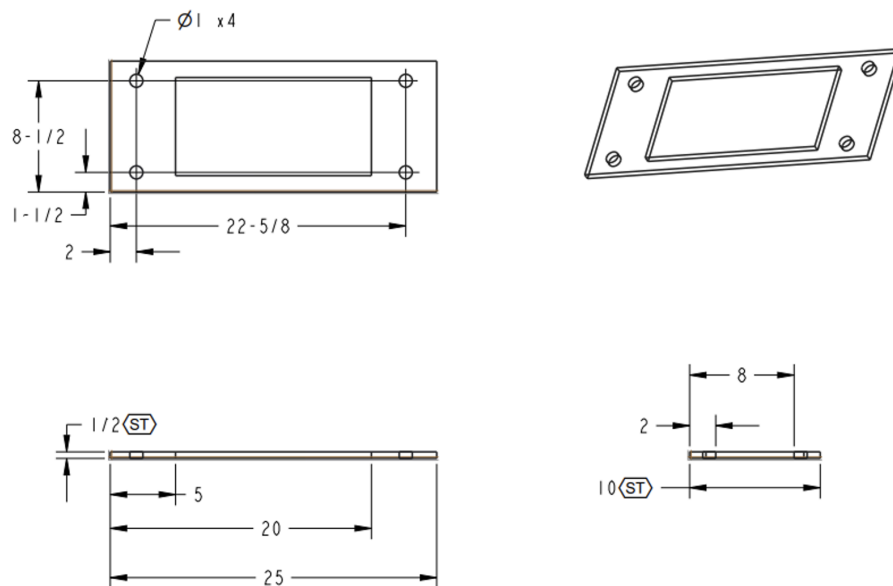


Figure 6: Base Plate Drawing

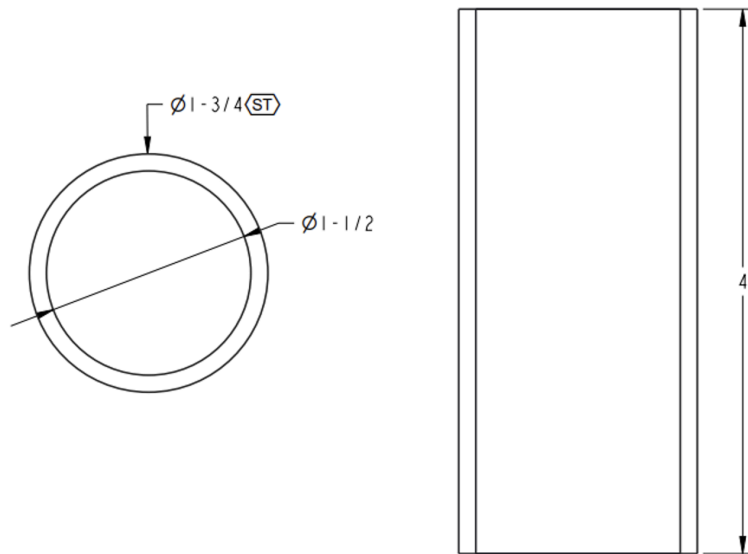


Figure 7: End Pipe Drawing

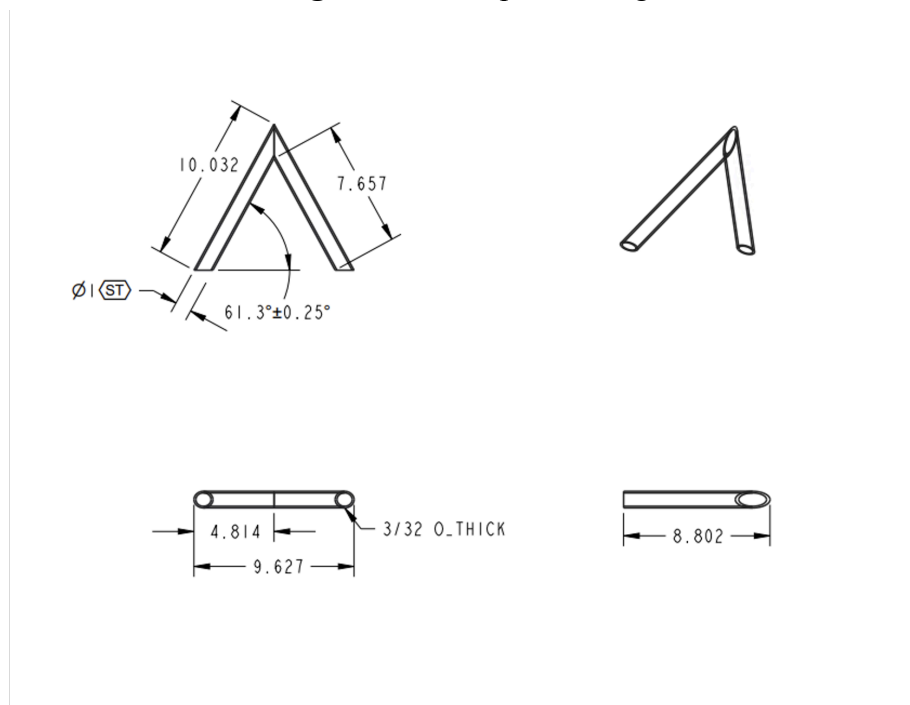


Figure 8: Triangular Hold Drawing